Structure-Awareness SAR Imagery by Exploiting Structure Tensor TV Regularization Under Multitask Learning Framework

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***Abstract*— Conventional sparsity-driven synthetic aperture radar (SAR) imagery often encounters the problem of loss of structural features in weak scattering. Although there are algorithms that focus on structure enhancement, no proper balance between accuracy and efficiency can be achieved. In this article, a novel feature enhancement algorithm, named structure-awareness SAR (SA-SAR), is proposed by exploiting an emerging regularizer of structure tensor total variation (STV). By imposing the STV norm onto the prior of the scenes or targets to be imaged, the intended structure feature can be analytically solved under the proximal algorithm. More specifically, the regularization method is developed within the alternating direction method of multipliers (ADMM) framework, where closed-form proximity operators can be derived. Due to the ADMM framework, it facilitates to incorporate with more features to be enhanced in a fully synergistic way. Therefore, the ℓ_1 and entropy norms are involved so that the sparse and focusing features can be enhanced accordingly. Considering the coherence of the SAR data, a linear proximal operator is developed within the multitask learning framework. The unavoidable error propagations can be alleviated in a great extent. In such cases, the proposed algorithm is superior in terms of convergence and efficiency. To facilitate the implementation and computation, the STV proximal mapping is optimized under the Vieta theorem. Finally, both simulated and raw SAR data are applied to verify the effectiveness of the proposed algorithm. Comparisons with conventional algorithms are carried out to show the superiority of the proposed algorithm.**

***Index Terms*— Alternating direction method of multipliers (ADMM), proximity operator, structure tensor total variation (STV), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a kind of all day-and-night and all weather remote sensing tool, which is characterized by the use of a physical antenna with limited aperture to achieve a large scale of synthesis aperture for a high-resolution and long-range imagery [1]–[6]. When SAR is mounted on an airborne platform, high-resolution images of ground scenes and moving targets can be obtained, which are usually referred to as SAR and ground moving target imagery (GMTIm) [7], [8]. While the radar is stationary on ground, it can be applied to image air-targets, which is named by inverse SAR (ISAR) [9], [10]. Conventional SAR image formation algorithms [8], [11] are mainly based on Fourier transformation. It has a common shortcoming that the resolution is limited by the transmitting bandwidth and the relative movement between the radar and the target of interest. Key features of the target become difficult to extract, and the accuracy for the target detection/recognition would be low.

Followed by the development of compressive sensing (CS) [12], [13], sparsity-driven SAR imagery becomes popular since it is capable of enhancing the dominant scattering feature of the target. By imposing a sparse prior onto the target to be imaged, a high-resolution SAR image can be recovered even with limited measurements and in a noisy environment. Generally, the sparsity-driven SAR imagery can be implemented in a greedy manner, Bayesian inference, and convex optimization [14], [15]. The last two categories can be viewed as the sparse prior introduced in statistical and regularized manners, respectively [16], [17]. Various methods have been proposed to improve the accuracy and/or promote the efficiency for the sparse feature recovery. However, followed by the enhanced sparse feature, the dominating scatterers are highlighted, while the weakness scattering structures would be easily lost. Therefore, it becomes necessary to explore detailed features of the target of interest.

To this end, SAR imaging algorithms with structural feature enhancement are to be investigated. Under sparse Bayesian learning (SBL) framework [18], a spatially continuous prior is introduced in addition to the sparse Laplacian prior [19], and both the sparse and structural features can be highlighted in the Bayesian posterior manner. Due to the large computational load, the practicability of SBL is limited, where only experiments in a small scale of data volume have been published. Although pattern-coupled SBL [20] is proposed, which can enhance the clustering feature in an improved efficiency, approximations from the solution of the approximated message passing (AMP) result in accuracy degradation. Alternatively, the structural feature can be strengthened in a regularization

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manner. By introducing ℓ_1/ℓ_2-mixed norm in the conventional convex optimization, the classical sparse Lasso model can be improved into the grouped Lasso [21], and the clustering structure can be highlighted with high efficiency. However, it is difficult for the grouped Lasso to capture complicated structures of the target prior, where too much manual interventions would decrease the automation of the algorithm.

Total variation (TV) [22] as an effective regularizer to formulate target features is becoming popular in recent years. TV regularization has been applied for SAR imagery in [23], where the target edge feature is highlighted and the speckle noise is suppressed. At the same time, it has the problems that the gradient transformation in the TV computation can only be obtained in two directions, which results in step artifacts and oversmoothing inhomogeneous regions [24]. To this end, many TV variants have been designed, which paves the way to formulate many useful features. Total generalized variation (TGV) [25] of order-h can encode high-order variation of structural features. Making h equal to 1, TGV becomes TV. To further accommodate structural features in an arbitrary direction, directional total variation (DTV) is developed in [26]. Although the TV variants are capable of overcoming drawbacks of the original TV, neighborhood information is not considered. Therefore, complicated and underlying structures are still difficult to be achieved. Structure tensor TV (STV) [27] is capable of utilizing structure tensor to capture the underlying attributes of the target to be imaged, where the adjacent scatterers can play a key role for the structural prior. However, the conventional STV definition involves complicated matrix eigenvalue decomposition [27], where the computational load is dramatically large, and the practicability would be limited.

Furthermore, for an accurate SAR imagery, there would be multiple features to be enhanced. Therefore, it is necessary to incorporate multiple priors under a cooperative framework. Alternating direction method of multipliers (ADMM) [28] is a simple but powerful tool in terms of multifeature enhancement [29]. Due to the variable splitting mechanism within the ADMM framework, multiple regularizers can be incorporated. The ADMM performs in a "Decomposition-Coordination" procedure, which ensures a satisfying convergence, and a balanced performance between accuracy and efficiency can be guaranteed [29].

In this article, to enhance structural features of the scenes or targets of interests, a novel structure-awareness SAR (SA-SAR) imaging algorithm is proposed, which exploits the STV norm to characterize the clustering feature of the target's scatterers. Specifically, the STV regularizer is optimized under the ADMM framework, which is capable of incorporating more priors to achieve an accurate SAR imagery. Different from the conventional STV computation, the proposed algorithm adopts determinant and trace of a matrix to replace the matrix eigenvalue decomposition under the Vieta theorem, which greatly reduces the computational complexity of the algorithm. Therefore, the STV regularizer can easily make use of the structure tensor to capture the first-order information of the scatterers of neighborhoods. Besides, the complex-valued data of radar echo are considered, and the STV regularization under ADMM is capable of accommodating for phase history of the SAR echoes. Proximal algorithm [30] is employed in solving the regularization problem, and an analytical proximity operator can be derived, which further simplifies the computation. Furthermore, to enhance the capability of autocalibration of both additive and multiplicative errors, the ℓ_1 norm and entropy norm are introduced into the ADMM framework. Super-resolution performance can therefore be ensured in limited measurements and low signal-to-noise ratio (SNR). Also, the possible phase errors in airborne SAR and GMTIm scenarios can be autocalibrated, and a satisfied focusing performance can be guaranteed. Due to the ADMM framework, the three regularizers are optimized in a collaborative multitask learning manner. Therefore, a superior convergence can be guaranteed, and the error propagation can be alleviated to a great extent. All the solutions are developed in analytical proximal operators, and high computational convenience can be ensured. Both simulated and raw SAR data are applied for the validation of the proposed algorithm. Comparisons with conventional algorithms are performed and evaluated in both qualitative and quantitative aspects.

This article is organized as follows. A generalized SAR/GMTIm signal model is given in Section II, and some preparations are introduced in Section III. In Section IV, the intended SA-SAR imaging algorithm is explained in detail. In Section V, the processing flowchart of the SA-SAR is shown. Both simulation and raw SAR data experiments are presented in Section VI. Finally, conclusions are drawn in Section VII.

II. Generalized SAR Signal Model

Generally, SAR transmits wideband pulses to achieve a high resolution in range direction while utilizing relative movements between the radar and target/scene to be imaged to form a virtual and synthesized aperture for a high resolution in azimuthal direction [31]. Commonly, it is assumed that the SAR mounted on an airborne platform moves along a predetermined flight track and the target is stationary or moving with linear trajectory. This can be referred to as SAR and GMTIm, respectively. Equivalently, when the radar is stationary on the ground, while the air-target is observed, it is the ISAR imagery. Therefore, the SAR and ISAR imagery can be equivalently transformed into the turntable model [29] after the translational movement is corrected, and the GMTIm can be viewed as a hybrid of SAR and ISAR imagery [29].

In Fig. 1, we assume $\mathbf{q}_0$ to be the reference position of the antenna phase center (APC) of the airborne SAR. In ideal cases, the airborne platform moves in a constant velocity v and a fixed altitude H, and $\mathbf{q}_0 = v t_n \vec{\mathbf{x}}$, where t_n is the slow-time variable under the "Stop-Go" assumption [32], and $\vec{\mathbf{x}}$ is the unit vector of the azimuthal direction. To accommodate for GMTIm, a multichannel SAR is required, where $\mathbf{q}_i = \mathbf{q}_0 + \mathbf{d}_i$ is used to denote the equivalent APC positions of the ith channel and $\mathbf{d}_i$ is the spacing vector from the reference channel. Next, the position of the pth stationary scatterer $\mathbf{r}_s^p(t_n)$ and the lth moving target $\mathbf{r}_t^l(t_n)$ is considered to be imaged. Defining $\mathbf{R}_0$ to be the reference range vector from the radar to the

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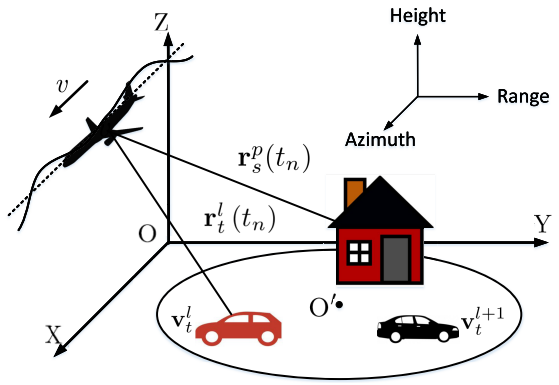


Fig. 1. Generalized SAR and GMTIm geometry.

observing scene center O′, $\mathbf{r}_0^p$ and $\mathbf{r}_0^l$ are the position offsets of the stationary scatterer and moving target from the scene center, respectively. Then, considering the moving target is with linear trajectory, the positions of the scatterer and target can be given as $\mathbf{r}_s^p(t_n) = \mathbf{R}_0 + \mathbf{r}_0^p$ and $\mathbf{r}_t^l(t_n) = \mathbf{R}_0 + \mathbf{r}_0^l + \mathbf{v}_t^l t_n$, respectively, where $\mathbf{v}_t^l$ is the moving target velocity. When the target is maneuvering, the acceleration of the target needs to be considered, which has been extensively investigated in our previous study [8], [18]. Under the above definitions, the ideal range vector of the reference channel to the stationary scatterer and moving target can be written as

$$\begin{cases} \mathbf{R}_s^p(t_n) = \mathbf{r}_s^p(t_n) - \mathbf{q}_0(t_n) \\ \mathbf{R}_t^l(t_n) = \mathbf{r}_t^l(t_n) - \mathbf{q}_0(t_n) \end{cases} \tag{1}$$

respectively. In practices, due to the atmospheric turbulence and/or navigation errors, the actual trajectory of either flying radar or moving target would deviate from the predetermination. Therefore, there would be range errors in the actual range history [18]. Defining $\Delta\mathbf{r}(t_n)$ to be the error vector, the actual range history for the stationary scatterer and moving target can be given as

$$\begin{cases} \tilde{R}_s^p(t_n) = \left|\mathbf{r}_s^p(t_n) - \mathbf{q}_0(t_n) - \Delta\mathbf{r}(t_n)\right| \\ \qquad\approx R_s^p(t_n) - \Delta r(t_n) \\ \tilde{R}_t^l(t_n) = \left|\mathbf{r}_t^l(t_n) - \mathbf{q}_0(t_n) - \Delta\mathbf{r}(t_n)\right| \\ \qquad\approx R_t^l(t_n) - \Delta r(t_n) \end{cases} \tag{2}$$

respectively, where $|\cdot|$ represents the vector module or the Euclidean norm, $R_s^p(t_n)$ and $R_t^l(t_n)$ denote the module of $\mathbf{R}_s^p(t_n)$ and $\mathbf{R}_t^l(t_n)$, respectively, which are the range history of stationary scatterer and moving target, respectively, under the ideal trajectory assumption, and $\Delta r(t_n)$ denotes the error deviation of $\Delta\mathbf{r}(t_n)$ projected onto the radar line-of-sight (LOS) [29].

Considering the linear frequency modulated (LFM) form of the transmitted pulses, the echoed spectra of the stationary scatterers and moving targets can be expressed as

$$\begin{cases} \tilde{S}_{s0}(k,t_n) = P(k)\sum_p \xi_p \exp\left[-jkR_s^p(t_n)\right]\exp[jk\Delta r(t_n)] \\ \qquad\qquad + n_s(k,t_n) \\ \tilde{S}_{t0}(k,t_n) = P(k)\sum_l \eta_l \exp\left[-jkR_t^l(t_n)\right]\exp[jk\Delta r(t_n)] \\ \qquad\qquad + n_t(k,t_n) \end{cases} \tag{3}$$

respectively, where $n(k,t_n)$ is the systematic thermal noise. In (3), we formulate the signal model in spatial frequency domain, where $k = (4\pi f)/c$ denotes the wavenumber variable, f gives the transmitted frequency, $f = f_0 + \Delta f$, f_0 is the center frequency, Δf is the frequency variation within the bandwidth, and c is the speed of light. It can be noted from (3) that the echoes from stationary scatterers consist of the transmitted signal modulation $P(k)$, the phase modulation from the systematic range history $R_s^p(t_n)$, and that from the nonsystematic range error of the flying radar. Besides, the echoes from moving targets consist of $P(k)$, the phase modulation from the systematic range history $R_t^l(t_n)$, and that from the nonsystematic range errors of the motion deviations of flying radar and moving target. Moreover, ξ_p and η_l in (3) denote the backscattering coefficient of the stationary scatterer and moving target, respectively.

Evaluating the systematic range history $R_s^p(t_n)$ and $R_t^l(t_n)$ in Taylor series expansion with respect to $|\mathbf{R}_0 - \mathbf{q}_0(t_n)|$ up to second order, there are

$$\begin{cases} R_s^p(t_n) = \left|\mathbf{R}_0 + \mathbf{r}_0^p - \mathbf{q}_0(t_n)\right| \\ \qquad = \left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right| + \dfrac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{\left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right|}\mathbf{r}_0^p + O\left(\mathbf{r}_0^p\right) \\ R_t^l(t_n) = \left|\mathbf{R}_0 + \mathbf{r}_0^l + \mathbf{v}_t^l t_n - \mathbf{q}_0(t_n)\right| \\ \qquad = \left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right| + \dfrac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{\left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right|}\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right) \\ \qquad\quad + \dfrac{\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right)^2}{2\left|\mathbf{R}_0 - \mathbf{q}_0(t_n)\right|} + O\left(\mathbf{r}_0^l + \mathbf{v}_t^l t_n\right) \end{cases} \tag{4}$$

where $O(\cdot)$ represents high-order Taylor series, which can be ignored under far-field assumption [18]. Let $\vec{\mathbf{r}}$ be the unit vector of slant range, and substitute $\mathbf{R}_0 = R_0\vec{\mathbf{r}}$, $\mathbf{r}_0^p = x^p\vec{\mathbf{x}} + r^p\vec{\mathbf{r}}$, $\mathbf{r}_0^l = x^l\vec{\mathbf{x}} + r^l\vec{\mathbf{r}}$, and $\mathbf{v}_t^l = v_x^l\vec{\mathbf{x}} + v_r^l\vec{\mathbf{r}}$ into (3), where x^p and r^p denote the azimuthal and slant-range positions of the pth stationary scatterer, respectively, x^l and r^l are that of the lth moving target, and v_x^l and v_r^l are the moving velocities in the azimuthal and slant-range directions, respectively. After the range deramping of $P(k)$, the echoed signal can be explicitly written as

$$\begin{cases} \tilde{S}_{s0}(k,t_n) = \sum_p \exp\left[-jk\left(\dfrac{R_0 r^p - v t_n x^p}{\sqrt{R_0^2 + (vt_n)^2}}\right)\right]\exp[j\Phi(k,t_n)] \\ \qquad\qquad + n_s(k,t_n) \\ \tilde{S}_{t0}(k,t_n) = \sum_l \exp\left[-jk\left(\dfrac{R_0 r^l - v t_n x^l}{\sqrt{R_0^2 + (vt_n)^2}}\right.\right. \\ \qquad\qquad + \dfrac{-vt_n v_x^l + \left(R_0 + r^l\right)v_r^l}{\sqrt{R_0^2 + (vt_n)^2}}t_n \\ \qquad\qquad \left.\left. + \dfrac{{v_x^l}^2 + {v_r^l}^2}{2\sqrt{R_0^2 + (vt_n)^2}}t_n^2\right)\right] \\ \qquad\qquad \cdot\exp[j\Psi(k,t_n)] + n_t(k,t_n) \end{cases} \tag{5}$$

where the multiplicative perturbation $\Phi(k,t_n) = \Psi(k,t_n) = k\Delta r(t_n)$ represents the phase error due to the radar and target motion deviations. By further noticing (5), the transmitting wavenumber k is coupled with azimuth time variable t_n, which results in systematic range migrations that will degrade the

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intended target image. According to the SAR image formation processing (IFP) [29], the classic polar format algorithm (PFA) [18] is employed to remove the range migration. According to the PFA processing, two interpolators, $k_r = R_0\ k/(R_0^2 + (vt_n)^2)^{1/2}$, $k_x = -vt_n k/(R_0^2 + (vt_n)^2)^{1/2}$, will be performed to resample the data from the polar format (k, t_n) into the rectangle format (k_r, k_x). Under the PFA interpolators, we have $k_x = -v\tilde{t}_n k_0/R_0$, where $\tilde{t}_n$ is the resampled slow-time variable and $k_0 = 4\pi\, f_0/\mathrm{c}$ is the wavenumber centroid. Then, the signal model in (5) can be expressed in the PFA resampled data domain as

$$\begin{cases} \tilde{S}_{s0}(k_x, k_r) = \sum_p \exp\left[-j\left(k_r r^p + k_x x^p\right)\right] \exp[j\Phi(k_x, k_r)] \\ \qquad\qquad + n_s(k_x, k_r) \\ \tilde{S}_{t0}(k_x, k_r) \\ \quad = \sum_l \exp\left[-j\left(k_r r^l + k_x x^l + \dfrac{k_0\left(R_0 + r^l\right)v_r^l}{R_0}\tilde{t}_n \right.\right. \\ \qquad\qquad \left.\left. + k_0\left(\dfrac{\left(v - v_x^l\right)^2 + {v_r^l}^2}{2R_0} - \dfrac{v^2}{2R_0}\right)\tilde{t}_n^2\right)\right] \\ \qquad\qquad \cdot \exp[j\Psi(k_x, k_r)] + n_s(k_x, k_r). \end{cases} \tag{6}$$

In (6), the echoed signal from the stationary scene is modeled in the first equation. While for the moving target, the echoed signal in the second equation can be generally obtained by the multichannel clutter suppression technique [18]. Furthermore, due to the target movement, the moving target signal is with additional modulations that are with respect to $\tilde{t}_n$ and $\tilde{t}_n^2$. In other words, both Doppler frequency shift and chirp rate are induced due to the moving target [18].

By observing (6), a uniform and compact form of signal model can be written in matrix expression as

$$\mathbf{Y} = \mathbf{EAX} + \mathbf{N} \tag{7}$$

where $\mathbf{N} \in \mathbb{C}^{N\times M}$ is the noise matrix, $\mathbf{Y} \in \mathbb{C}^{N\times M}$ denotes the echo signal in range-compressed domain of (6), and N and M represent the number of echoed pulses and range cells, respectively. In (7), $\mathbf{E}$ is a diagonal matrix, where the diagonal elements correspond to the value of the phase error at different times, which can be expressed as $[e^{\mathrm{j}\phi(\tilde{t}_1)}, \ldots, e^{\mathrm{j}\phi(\tilde{t}_N)}]$, and $\phi(\tilde{t}_n)$ is the nonsystematic error from the motion deviations of airborne radar and moving targets, as Φ and Ψ shown in (6). Furthermore, $\mathbf{A}$ in (7) is the dictionary for the inversion of $\mathbf{X}$, which can be generally given as a Fourier transform matrix $\mathbf{A}_0$ that is $\mathbf{A} = \mathbf{A_0} = [\mathbf{a}(\hat{\mathrm{f}}_\mathbf{d}^\mathbf{1}), \ldots, \mathbf{a}(\hat{\mathrm{f}}_\mathbf{d}^\mathbf{N})]$, $\mathbf{a}(\hat{f}_d^n) = [e^{-j2\pi \hat{f}_d^n \tilde{t}_1}, \ldots, e^{-j2\pi \hat{f}_d^n \tilde{t}_N}]^{\mathrm{T}}$, where $\hat{f}_d^n$ indicates the Doppler-frequency index and $[\cdot]^{\mathrm{T}}$ gives the vector/matrix transpose operator. When moving targets are concerned, $\mathbf{A}$ should be a parametric combination of quadratic Fourier dictionary as $\mathbf{A} = [\mathbf{A}_1, \ldots, \mathbf{A}_L]$, where each element $\mathbf{A}_l$ is to focus the lth target. Specifically, $\mathbf{A}_l = \mathbf{A}_0 \odot \mathbf{D}_l$, where $\odot$ denotes element-wise Hadamard product operator, $\mathbf{D}_l = [\mathbf{D}_l(\gamma_d^l), \ldots, \mathbf{D}_l(\gamma_d^l)]_{1\times N}$, $\mathbf{D}_l(\gamma_d^l) = [e^{j\pi\gamma_d^l t_1^2}, \ldots, e^{j\pi\gamma_d^l t_N^2}]^{\mathrm{T}}$, and $\boldsymbol{\gamma}_d = \{\gamma_d^l\}_L$ gives the Doppler chirp rates of the L moving targets to be imaged. Adopting LVD [8] for the time–frequency representation of multiple targets, $\boldsymbol{\gamma}_d$ can be obtained. Finally, $\mathbf{X}$ is the intended SAR image of either stationary scenario or moving target [29]. When L moving targets are concerned, the solution should be $\sum_{l=1}^{L} \mathbf{X}_{[(l-1)N+1:lN],:}$.

III. Preparations

Before we are going to discuss the intended SA-SAR imaging algorithm, some preparations are given for fundamentals.

A. Regularization

As shown in (7), to solve the inverse problem for $\mathbf{X}$, a simple matrix inversion as $\mathbf{A}^{-1}$ can be used. However, due to the additive noise and possible singular matrix $\mathbf{A}$, the inverse problem would become ill-conditioned, and no proper solution can be guaranteed [29]. Commonly, optimization with a proper regularizer is capable of overcoming the difficulty, where the objective function in an unconstrained form can be written as

$$\zeta(\mathbf{X}) = \frac{1}{2}\|\mathbf{Y} - \mathbf{EAX}\|_{\mathrm{F}}^2 + \xi\, v(\mathbf{X}) \tag{8}$$

where $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm, and it is the fidelity term denoting the accuracy for the inverse problem of (7). $v(\mathbf{X})$ in (8) is the intended regularizer for a proper prior, and ξ is the regularization parameter. The regularizer plays an important role to constrain the optimization processing for an intended solution. Therefore, the regularization optimization is capable of incorporating prior knowledge to recover the intended signal/image.

B. Structure Tensor Total Variation

Regarding the optimization regularizer, TV norm is a popular choice, which is capable of preserving outlines of the target structures. Mathematically, the TV norm is defined as

$$\mathrm{TV}(\mathbf{X}) = \int_\Omega \|\nabla\mathbf{X}(x)\|_2 dx \tag{9}$$

where $\mathbf{X}$ is the target image, x denotes the specific resolution cell, Ω is the image space for the integral domain, $\|\cdot\|_2$ is the ℓ_2-norm, and ∇ is the gradient operator. It can be noted from (9) that the TV norm is the sum of the two directional gradients of all resolution cells in the target image. However, the TV norm only penalizes the differences in two directions. Therefore, it often encounters the problem of oversmoothing and staircase artifacts, which would limit its application in complicated target structure scenario.

To this end, a novel TV variant with the structure tensor formulation, or named by STV [27], is introduced. STV generally captures the first-order information around a local neighborhood of scatterers to measure the local variation. In such a case, structural details can be captured and modeled, which makes it easy to establish priors for the target with complicated features. The structure tensor of the target scenario $\mathbf{X}$ at any cell x is given by

$$S_G\mathbf{X}(x) = G * \left[\nabla\mathbf{X}(x)\nabla\mathbf{X}(x)^{\mathrm{H}}\right] \tag{10}$$

where G is a Gaussian kernel used for weighted smoothing [33] and $*$ is the convolution operator.

Next, under the STV definition, using the eigenvalue decomposition in the structure tensor, it is capable of sensing

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complicated features of the target to be imaged. For scatterers distributed in point-like, curved lines, or nonhomogeneous surface, they all can be characterized and formulated by the combinations of eigenvalues and eigenvectors. Specifically, the STV norm can be defined by the eigenvalues of the structure tensor as

$$\mathrm{STV}_p(\mathbf{X}) = \int_{\Omega} \left(\sum_i \lambda_i^{p/2}(x) \right)^{1/p} dx \tag{11}$$

where λ_i is the ith eigenvalue of the structure tensor of $S_G\mathbf{X}$, p is an arbitrary constant, and generally, $p \geq 1$.

C. Alternating Direction Method of Multipliers

In conventional regularization optimizations, only a single regularizer is incorporated, which will accordingly enhance the feature as the prior modeled. However, a single regularizer would be insufficient to capture the feature of the real scenario. There will be diverse features in the high-resolution SAR imagery, where multiple regularizers are required to formulate the rich prior content [29], [34].

ADMM [35] is an emerging method that has strong potentials for multiple-feature enhancement. Inherently, ADMM is a combined methods of dual ascent and method of multipliers [29], which improves the convergence of the iteration. What is more important, the variable splitting mechanism in ADMM can be exploited to decompose a joint optimization task into a series of simple and convex subtasks, and they can be designed to be carried out in a collaborative manner. In this article, a multitask learning framework is established under ADMM, and the intended SA-SAR algorithm is proposed under the multitask learning framework. Considering the problem of multiple-feature enhancement in a constrained form as

$$\begin{aligned} &\arg\min_{\mathbf{X}} \sum_{g=1}^{G} \xi_g \Theta_g(\mathbf{X}) \\ &\text{subject to } \|\mathbf{Y} - \mathbf{EAX}\|_{\mathrm{F}}^2 \leq \varepsilon \end{aligned} \tag{12}$$

where G represents that there are G features to be enhanced, $\Theta_g(\mathbf{X})$ is the regularizer for the gth feature, ξ_g is the regularization parameter of the gth regularizer, and ε is related to the noise power. According to the variable splitting mechanism in the ADMM, (12) can be equivalently expressed as

$$\underset{\mathbf{X},\mathbf{Z}}{\text{minimize}}\ f(\mathbf{X}) + g(\mathbf{Z}), \quad \text{s.t. } \mathbf{KX} - \mathbf{PZ} = \mathbf{S} \tag{13}$$

where $f(\mathbf{X}) = (1/2)\|\mathbf{EAX} - \mathbf{Y}\|_{\mathrm{F}}^2$ is the data fidelity, the splitted variable $\mathbf{Z}$ is introduced for $g(\mathbf{Z}) = \sum_G \xi_g \Theta_g(\mathbf{Z}_g)$ representing a joint objection function for the feature enhancement, $\mathbf{Z} = [\mathbf{Z}_1, \ldots, \mathbf{Z}_G]^{\mathrm{T}}$ is a set of split variables corresponding to G features, and $\mathbf{K}$, $\mathbf{P}$, and $\mathbf{S}$ are the coefficient matrices, which constrain the equivalent relationship between $\mathbf{X}$ and $\mathbf{Z}$. Setting $\mathbf{K} = [\mathbf{K_1}, \ldots, \mathbf{K_G}]^{\mathrm{T}}$, $\mathbf{P} = \mathbf{I}$ and $\mathbf{S} = \mathbf{0}$, we have $\mathbf{Z}_g = \mathbf{K}_g\mathbf{X}$. Then, the unconstrained function in augmented Lagrangian form can be expressed as

$$\begin{aligned} L(\mathbf{X}, \mathbf{Z}; \mathbf{U}) = f(\mathbf{X}) + g(\mathbf{Z}) + \mathbf{U}^{\mathrm{H}}(\mathbf{KX} - \mathbf{PZ} - \mathbf{S}) \\ + \frac{\rho}{2}\|\mathbf{KX} - \mathbf{PZ} - \mathbf{S}\|_2^2 \end{aligned} \tag{14}$$

where $\mathbf{U}$ is the dual variable, $\{\cdot\}^{\mathrm{H}}$ is the conjugate transpose operator, and ρ is the augmented Lagrangian parameter. To solve the problem, coordinate descent scheme can be adopted [29], [36], and the procedure can be written in mathematics as

$$\begin{cases} \text{for } g = 1 : G \\ \quad \mathbf{Z}_g^{k+1} = \arg\min_{\mathbf{Z}_g} \left\{ \xi_g \Theta_g(\mathbf{Z}_g) + \frac{\rho}{2} \left\| \mathbf{K}_g\mathbf{X}^k - \mathbf{PZ}_g - \mathbf{D}_g^k \right\|_{\mathrm{F}}^2 \right\} \\ \quad \mathbf{D}_g^{k+1} = \mathbf{D}_g^k - \mathbf{K}_g\mathbf{X}^k + \mathbf{PZ}_g^{k+1} \\ \text{end} \\ \mathbf{Z}^{k+1} = \left[\mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1}, \ldots, \mathbf{Z}_G^{k+1} \right] \\ \mathbf{D}^{k+1} = \left[\mathbf{D}_1^{k+1}, \mathbf{D}_2^{k+1}, \ldots, \mathbf{D}_G^{k+1} \right] \\ \mathbf{X}^{k+1} = \arg\min_{\mathbf{X}} \left\{ f(\mathbf{X}) + \frac{\rho}{2} \left\| \mathbf{KX} - \mathbf{PZ}^{k+1} - \mathbf{D}^{k+1} \right\|_{\mathrm{F}}^2 \right\} \end{cases} \tag{15}$$

where the superscript k represents the iteration number and $\mathbf{D} = (\mathbf{U}/\rho)$ is the scaled dual variable. As it can be noted from (15), under the ADMM framework, the joint feature enhancement problem is decomposed into G subproblems, and they can be optimized in a decentralized manner. After the "for-end" circulation, all $\mathbf{Z}_g^{k+1}$ and $\mathbf{D}_g^{k+1}$ can be collected and gathered into $\mathbf{Z}^{k+1}$ and $\mathbf{D}^{k+1}$. In such cases, the final $\mathbf{X}^{k+1}$ can be updated accordingly. Due to the "Decomposition-Coordination" procedure, the ADMM framework has great potential in achieving a true multitask learning.

IV. SA-SAR Imaging Algorithm

Under the preparations mentioned above, a novel SA-SAR imaging algorithm will be proposed in this section, which mainly utilizes the STV regularization under the multitask learning framework. To further incorporate capabilities of super-resolution and error autocalibration, the ℓ_1 norm and entropy norm are involved in the ADMM framework. Therefore, three splitted variables $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ are required, and totally, three features ($G = 3$) are to be formulated. In such cases, the joint feature enhancement problem in the unconstrained form can be given as

$$\begin{aligned} \hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{EAX}\|_2^2 + \xi_1\|\mathbf{Z}_1\|_{\mathrm{STV}} \\ + \xi_2\|\mathbf{Z}_2\|_{\mathrm{EN}}^2 + \xi_3\|\mathbf{Z}_3\|_1 \end{aligned} \tag{16}$$

where three regularizers are incorporated and $\|\cdot\|_{\mathrm{STV}}$, $\|\cdot\|_{\mathrm{EN}}$, and $\|\cdot\|_1$ are the STV norm, entropy norm, and ℓ_1 norm, respectively. They are to formulate the structural feature, focusing feature, and sparse feature, respectively. In (16), ξ_1, ξ_2, and ξ_3 are the regularization parameters for the specific regularizer. More specifically, if $\mathbf{K} = [\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3]$, $\mathbf{P} = \mathbf{I}$, and $\mathbf{S} = \mathbf{0}$, we have the equalities $\mathbf{Z}_1 = \mathbf{K}_1\mathbf{X}$, $\mathbf{Z}_2 = \mathbf{K}_2\mathbf{X}$, and $\mathbf{Z}_3 = \mathbf{K}_3\mathbf{X}$. Each constant matrix $\mathbf{K}_g$ will be fixed in the following discussion. Next, according to the calculation procedure as in (15), each splitted variable or feature variable $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ is to be derived. Furthermore, proximal algorithm [30], [37] is employed to ensure closed-form solutions for the feature proximity operators.

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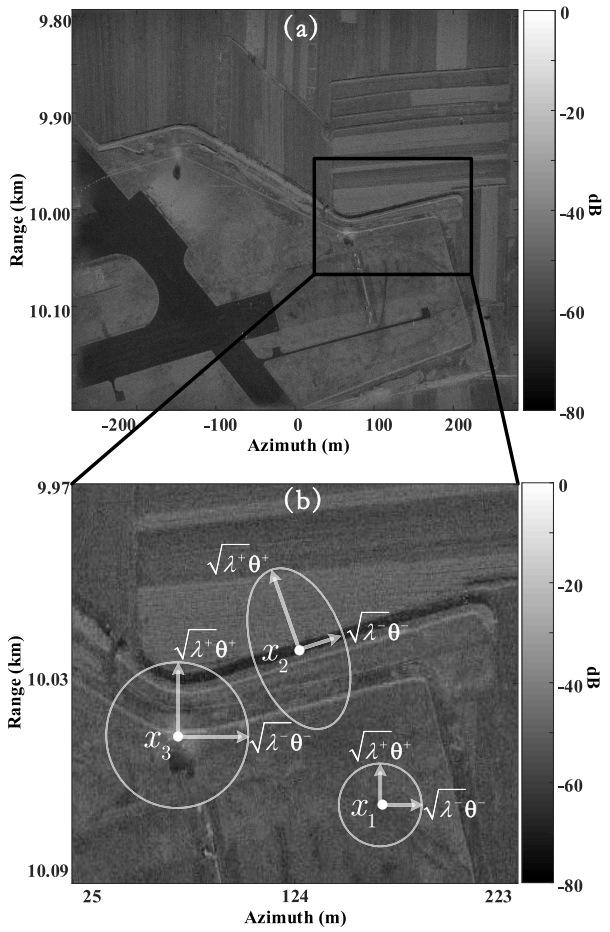


Fig. 2. Schematic of multiple structural features in SAR images.

A. Local Optimization #1: Structure Feature Enhancement

To highlight structural details of the target of interest during the SAR imagery, a novel STV regularizer is employed. Compared with the conventional STV regularizer, the intended STV norm is easier and more efficient to compute, and it is convenient to incorporate within the ADMM framework. Regarding the conventional STV norm, it can be noted from (11) that the eigenvalue decomposition is required at each resolution cell. Considering the 2-D SAR imagery, there are two eigenvalues λ_x^+ and λ_x^- to be achieved, where $\lambda_x^+ \geq \lambda_x^- \geq 0$. Under the eigenvalues and corresponding eigenvectors, scatterers' clustering features can be represented. As shown in Fig. 2, there is a typical scenario to be imaged by SAR, where diverse features are to be enhanced. For smoothed clustering features, e.g., the plane surface in Fig. 2(b) (x_1), the two eigenvalues are both small and with closed value, where θ^+ and θ^- are orthotropic eigenvectors. On the contrary, when the two eigenvalues are both large and with closed value [x_3 in Fig. 2(b)], the point-like feature can be indicated under the STV definition. Besides, when the values are different from each other [x_2 in Fig. 2(b)], the features of scatterers distributed in a curved line can be formulated.

In such cases, under the STV formulation, the point-like, curved-line, and plane-surface features can all be modeled mathematically. Great potentials can be exploited for the enhancement of detailed features. However, the direct definition of the STV norm in (11) is difficult to incorporate within the ADMM framework, where there is no distinct expression in variable splitting form. In addition, the eigenvalue decomposition is not suitable for large scale of data matrix, where the computational time will be consumed in an unacceptable level. To this end, a novel STV definition is introduced, which has a convenient form to compute, and it is easy to be involved within the intended multitask learning framework. Suppose that $\vec{\mathbf{n}} = (\vec{\mathbf{n}}_1, \vec{\mathbf{n}}_2)$ is an arbitrary direction, and $\vec{\mathbf{n}}_1$ and $\vec{\mathbf{n}}_2$ are the unit vectors in orthogonal directions. Then, the directional derivative of the target to be imaged $\mathbf{X}$ can be written at any specific resolution cell x as

$$\frac{\partial \mathbf{X}}{\partial \vec{\mathbf{n}}}(x) = [\mathbf{J}\mathbf{X}(x)]\vec{\mathbf{n}} \tag{17}$$

where $\mathbf{J}\mathbf{X}$ is the Jacobian matrix of $\mathbf{X}$, which is defined as

$$\mathbf{J}\mathbf{X}(x) = [\nabla \mathbf{X}_1(x) \cdots \nabla \mathbf{X}_I(x)]^{\mathrm{T}} \tag{18}$$

where I denotes for the number of data channels for the specific data format. For instance, $I = 1$ is for the real-valued data and $I = 2$ is for the complex-valued data with real and imaginary parts. Due to the Jacobian matrix, it is capable of accommodating for phase history data in complex valued [27], where an increased dimension of matrix computation will be involved. Therefore, it is possible for the STV regularization to be phase preservation. Next, the structure tensor in (10) can be rewritten in terms of Jacobian matrix as

$$S_G\mathbf{X}(x) = G * \left[\mathbf{J}\mathbf{X}^{\mathrm{H}}\mathbf{J}\mathbf{X}\right](x) = \mathbf{J}_G\mathbf{X}(x)^{\mathrm{H}}\mathbf{J}_G\mathbf{X}(x) \tag{19}$$

where G is the Gaussian kernel, $\mathbf{J}_G$ is the Gaussian-weighted Jacobian matrix, and the superscript H denotes the operator of matrix conjugate transpose. Compared with the conventional definition in (10), where the time-consuming matrix convolution is required, the new definition in (19) only involves matrix multiplication of Gaussian-weighted Jacobian matrix. More importantly, in (19), it reveals that the eigenvalues of the structure tensor $S_G\mathbf{X}(x)$ are equivalent to the singular values of $\mathbf{J}_G\mathbf{X}(x)$ [38]. Therefore, the weighted Jacobian matrix $\mathbf{J}_G$ paves a way to the coefficient matrix $\mathbf{K}_1$ for the regularization in the ADMM manner. It is capable of getting a closed-form expression for the proximal mapping of the STV regularizer.

Then, the novel STV norm can be written as

$$\mathrm{STV}_p(\mathbf{X}) = \sum_{x=1}^{MN} \left\| \left(\sqrt{\lambda_x^+}, \sqrt{\lambda_x^-} \right) \right\|_p = \sum_{x=1}^{MN} \|[\mathbf{J}_G\mathbf{X}]_x\|_{sp} \tag{20}$$

where s in $\|\cdot\|_{s,p}$ represents the singular value of the matrix $\mathbf{J}_G\mathbf{X}$ and p stands for p-norm calculation of the singular value. Furthermore, the summation term of the s, p-norm in (20) can be treated as ℓ_1 norm of the s, p-norm. Therefore, $\|\cdot\|_{1,p}$ is defined as the sum of $\|\cdot\|_{s,p}$ norm, and the intended STV norm can be defined as a $\ell_{1,p}$ mixed norm as

$$\mathrm{STV}_p(\mathbf{X}) = \sum_{x=1}^{MN} \|[\mathbf{J}_G\mathbf{X}]_x\|_{sp} = \|\mathbf{J}_G\mathbf{X}\|_{1,p}. \tag{21}$$

Compared with the conventional definition in (11), where the STV is the $\ell_{1,p}$ norm of eigenvalues of the structure tensor

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matrix at x, the new definition in (21) transfers the STV into a $\ell_{1,p}$ norm of a Jacobian matrix of $\mathbf{X}$ at x. In such cases, it becomes convenient to determine the STV regularization in the ADMM form as shown in (15). Setting the first splitted variable or the structural variable $\mathbf{Z}_1 = \mathbf{J}_G\mathbf{X}$, the structural feature enhancement can be explicitly given as

$$\mathbf{Z}_1^{k+1} = \arg\min_{\mathbf{Z}_1} \xi_1 \|\mathbf{Z}_1\|_{1,p} + \frac{\rho}{2}\left\|\mathbf{J}_G\mathbf{X}^k - \mathbf{Z}_1 - \mathbf{D}_1^k\right\|_{\mathrm{F}}^2 \quad (22)$$

where $\mathbf{J}_G$ is the coefficient matrix $\mathbf{K}_1 = \mathbf{J}_G$ in the variable splitting in (13). In (22), a typical minimization problem can be observed, and the proximal algorithm can be applied for a closed-form solution of the structural feature.

However, the calculation of the proximity operator involves either eigenvalue decomposition or singular decomposition. It needs large computational expenses. In this article, we explore an alternative calculation of STV, which only utilizes the trace and determinant of a matrix, and the computational complexity can be decreased in a great extent. Under the Vieta theorem of matrix fundamentals, the sum of the eigenvalues of an arbitrary matrix equals the trace of the matrix, and the product of the eigenvalues equals the matrix determinant. Therefore, we have $\lambda^+ + \lambda^- = \mathrm{T}$ and $\lambda^+ \cdot \lambda^- = \mathrm{H}$, where T and H are the trace and determinant of the matrix, respectively. When the ℓ_1, ℓ_2, and ℓ_∞ norms of the eigenvalues are written as

$$\begin{cases} \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_1 = \sqrt{\lambda^+} + \sqrt{\lambda^-} \\ \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_2 = \sqrt{\lambda^+ + \lambda^-} \\ \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_\infty = \sqrt{\lambda^+} \end{cases} \quad (23)$$

respectively, they can be further expressed as

$$\begin{cases} \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_1 = \sqrt{\mathrm{T} + 2\sqrt{\mathrm{H}}} \\ \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_2 = \sqrt{\dfrac{\left(\sqrt{\mathrm{T} + 2\sqrt{\mathrm{H}}}\right)^2 + \left(\sqrt{\mathrm{T} - 2\sqrt{\mathrm{H}}}\right)^2}{2}} \\ \qquad\qquad = \sqrt{\mathrm{T}} \\ \left\|\sqrt{\lambda^+}, \sqrt{\lambda^-}\right\|_\infty = \dfrac{\sqrt{\mathrm{T} + 2\sqrt{\mathrm{H}}} + \sqrt{\mathrm{T} - 2\sqrt{\mathrm{H}}}}{2} \end{cases} \quad (24)$$

respectively. Therefore, the calculation of the eigenvalues, which involves complicated matrix decomposition, can be equivalently transformed into the calculation of the matrix trace and determinant, where only simple matrix trace and determinant are to be computed. In accordance, the computational complexity can be reduced dramatically.

Under the definition of the STV norm in matrix trace and determinant manner, the formulation of the structural features in Fig. 2 can be interpreted in an alternative view. First, when $\mathrm{T} = 0$, both the positive eigenvalues would be small and close to each other. Therefore, the smoothing plane surface can be formulated as x_1 in Fig. 2. Then, if $\mathrm{T} \geq 0$ and $\mathrm{H} = 0$, it is equivalent to the situation that one of the eigenvalues is large, while the other one is small. In such cases, the features in curved lines can be captured as x_2 in Fig. 2. Finally, when $\mathrm{T} \geq 0$ and $\mathrm{H} \geq 0$, this means that both the eigenvalues are approximately equal and large, and the point-like feature can be modeled as x_3 in Fig. 2. It can be noted from the above analysis that using the matrix trace and determinant calculation can equivalently demonstrate structural features of the target to be imaged, while a decreased computational load improves the efficiency of the intended feature enhancement algorithm.

Focusing on the objective function of the optimization problem in (22), the STV regularizer is in the form of ℓ_1 norm. According to the proximal algorithm [37], [39], the proximity operator of the ℓ_1 norm regularizer can be expressed as a soft-thresholding operator. Therefore, the updating function for the structural feature variable $\mathbf{Z}_1$ at the $k+1$th iteration can be written as

$$\mathbf{Z}_1^{k+1} = \mathrm{prox}_{\frac{\xi_1}{\rho}\|\cdot\|_{1,p}}\left(\mathbf{J}_G{\mathbf{W}_1}^{k+1} - \mathbf{D}_1^k\right) \quad (25)$$

where $\mathrm{prox}(\cdot)$ is the soft-thresholding operator, and its specific expression can refer to [29]. The variable ${\mathbf{W}_1}^{k+1} = \mathbf{A}^{\mathrm{H}}\{(\mathbf{E}^k)^{\mathrm{H}}\mathbf{A}\mathbf{X}^k\} - \mathbf{D}_1^k$ gives the phase corrected data matrix to accommodate for the possible nonsystematic phase errors in $\mathbf{E}$, and $\mathbf{D}_1$ is the dual variable to measure the residual error for the variable splitting. From (25), to update $\mathbf{Z}_1$, the Jacobain matrix $\mathbf{J}_G$ should be determined previously, and the image matrix $\mathbf{X}$, the phase error estimation $\mathbf{E}$, and the dual variable $\mathbf{D}_1$ at the kth iteration are required.

Then, under the ADMM procedure in (15), the dual variable can be updated as

$$\mathbf{D}_1^{k+1} = \mathbf{D}_1^k - \mathbf{J}_G{\mathbf{W}_1}^{k+1} + \mathbf{Z}_1^{k+1}. \quad (26)$$

From the above developments, the structural feature $\mathbf{Z}_1$ and the related dual variable $\mathbf{D}_1$ are obtained in closed-form solution in (25) and (26), respectively. It can be viewed as a local optimization for the structure feature enhancement.

B. Local Optimization #2: Focusing Feature Enhancement

Next, to ensure the focusing quality of the SAR imagery in airborne scenario and/or moving target imagery, the nonsystematic phase error as modeled in $\mathbf{E}$ in (7) is to be investigated. At this stage, we are to embed the error autocalibration within the ADMM framework and realize the focusing feature enhancement in the second local optimization.

Introducing the entropy norm [29] to denote the focusing quality of SAR imagery, we can set the second splitted variable or feature variable $\mathbf{Z}_2$ equal to the primal variable $\mathbf{X}$, and $\mathbf{K}_2 = \mathbf{I}$. Therefore, the focusing feature enhancement can be formulated as

$$\mathbf{Z}_2^{k+1} = \arg\min_{\mathbf{Z}_2}\left\{-\frac{\xi_2}{\mathrm{E}_{\mathbf{Z}_2}}\sum_{n,m}|\mathbf{Z}_2(n,m)|^2 \ln|\mathbf{Z}_2(n,m)|^2 + \ln\mathrm{E}_{\mathbf{Z}_2} + \frac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_2 - \mathbf{D}_2^k\right\|_{\mathrm{F}}^2\right\} \quad (27)$$

where the regularizer in the first two terms is the definition of the entropy norm, ξ_2 is the regularization parameter, and $\mathrm{E}_{\mathbf{Z}_2}$ is the total energy of $\mathbf{Z}_2$. Furthermore, $\mathbf{Z}_2$ can be expressed in the error-calibrated form as

$$\mathbf{Z}_2(n,m) = \frac{1}{N}\sum_{q=0}^{N-1}\exp\left(\mathrm{j}2\pi\frac{qn}{N}\right)\exp\left(j\phi_q\right)\widetilde{\mathbf{Y}}_{qm} \quad (28)$$

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where $\widetilde{\mathbf{Y}}_{qm}$ is the range-compressed data and ϕ_q denotes the qth element of the vector of the phase error correction. Setting $\boldsymbol{\phi}$ to be the phase vector for the correction of the nonsystematic error and substituting (28) into (27), the objective function of the entropy norm regularization with respect to the phase vector $\boldsymbol{\phi}$ can be obtained. However, due to the logarithmic term in (27), the objective function is nonconvex, and there will be no closed-form solution. The convergence of the optimization will be degraded. To this end, a surrogate function for the original objective function is required. Supposing $\boldsymbol{\phi}_l$ to be the phase correction vector in a coordinate descent manner, where the lth element of $\boldsymbol{\phi}$ is to be estimated, while others are fixed with the estimated values, the criterion to determine the surrogate function can be shown as

$$\begin{cases} \Theta\big(\mathbf{Z}_2(\boldsymbol{\phi}_l);\mathbf{Z}_2(\boldsymbol{\phi}_l)\big) = \Theta(\mathbf{Z}_2(\boldsymbol{\phi}))|_{\boldsymbol{\phi}=\boldsymbol{\phi}_l} \\ \dfrac{\partial}{\partial\boldsymbol{\phi}}\Theta\big(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l)\big)\Big|_{\boldsymbol{\phi}=\boldsymbol{\phi}_l} = \dfrac{\partial}{\partial\boldsymbol{\phi}}\Theta(\mathbf{Z}_2(\boldsymbol{\phi}))\Big|_{\boldsymbol{\phi}=\boldsymbol{\phi}_l} \\ \Theta\big(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l)\big) \geq \Theta(\mathbf{Z}_2(\boldsymbol{\phi})) \end{cases} \tag{29}$$

where $\Theta(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l))$ is the intended surrogate function. It can be noted from (29) that the first two equations ensure that both the original and surrogate functions reach the same value at the same point, and the third equation guarantees that the surrogate always lies above the original function. Under the criterion in (29), the surrogate function can be designed as [29]

$$\begin{aligned} &\Theta\big(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l)\big) \\ &= -\frac{\xi_2}{\mathrm{E}_{\mathbf{Z}_2}}\sum_{n,m}|\mathbf{Z}_2(n,m;\boldsymbol{\phi})|^2\ln\big|\mathbf{Z}_2\big(n,m;\boldsymbol{\phi}_l\big)\big|^2 \\ &\quad + \ln\mathrm{E}_{\mathbf{Z}_2} + \frac{\rho}{2}\sum_{n,m}|\mathbf{Z}_2(n,m,\boldsymbol{\phi})|^2 + \frac{\rho}{2}\big\|\mathbf{X}^k - \mathbf{D}_2^k\big\|_{\mathrm{F}}^2 \end{aligned} \tag{30}$$

where the logarithmic term is fixed by the $\boldsymbol{\phi}_l$ and the function becomes convex. Therefore, both closed-form solutions and superior convergence can be guaranteed. Noticing the surrogate function in (30), we find $\Theta'(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l)) + \Theta'''(\mathbf{Z}_2(\boldsymbol{\phi});\mathbf{Z}_2(\boldsymbol{\phi}_l)) = 0$, where $\Theta'(\cdot)$ and $\Theta'''(\cdot)$ denote the first- and third-order derivatives of the surrogate function with respect to $\boldsymbol{\phi}$, respectively. Therefore, solving the problem of minimization of the surrogate function is inherently to solve a high-order differential equation. The updating equation can be accordingly written as [29]

$$\phi_l^{k+1} = \phi_l^k - \arctan\frac{\Theta_l''}{\Theta_l'} \tag{31}$$

where Θ_l' and Θ_l'' are the first- and second-order derivatives of the surrogate function with respect to the lth element of $\boldsymbol{\phi}$, respectively. When all the elements of ϕ are updated, the diagonal matrix $\mathbf{E}_{k+1}$, whose diagonal elements expressed as $[e^{\mathrm{j}\phi(\tilde{t}_1)},\ldots,e^{\mathrm{j}\phi(\tilde{t}_N)}]$, can be achieved, and the intended focusing feature variable $\mathbf{Z}_2$ can be solved as

$$\mathbf{Z}_2^{k+1} = \big(\mathbf{E}^{k+1}\big)^{\mathrm{H}}\mathbf{A}^{\mathrm{H}}\mathbf{Y}. \tag{32}$$

Similarly, the second dual variable $\mathbf{D}_2$ can be updated as

$$\mathbf{D}_2^{k+1} = \mathbf{D}_2^k - \mathbf{X}^{k+1} + \mathbf{Z}_2^{k+1}. \tag{33}$$

In such cases, the focusing feature enhancement is fulfilled, where the splitted variable and dual variable can be updated as (32) and (33), respectively.

C. Local Optimization #3: Sparse Feature Enhancement

Then, to guarantee the performance of super-resolution of SAR imagery, ℓ_1 is introduced as a regularizer for an additional prior, and the sparse feature can be enhanced accordingly. Specifically, from (15), setting $\mathbf{K}_3$ to be $\mathbf{I}$, the third splitted variable $\mathbf{Z}_3 = \mathbf{X}$, and the third local optimization can be expressed as

$$\mathbf{Z}_3^{k+1} = \arg\min_{\mathbf{Z}_3}\ \xi_3\left\|\mathrm{vec}(\mathbf{Z}_3)\right\|_1 + \frac{\rho}{2}\big\|\mathbf{X}^k - \mathbf{Z}_3 - \mathbf{D}_3^k\big\|_{\mathrm{F}}^2 \tag{34}$$

where $\mathrm{vec}(\cdot)$ denotes the vectorization of a matrix, $\|\cdot\|_1$ is the ℓ_1 norm operator, and ξ_3 is the corresponding regularization parameter. Due to the problem in (34) which is in a typical Lasso model, the solution for the sparse feature $\mathbf{Z}_3$ can be given in soft-thresholding form as

$$\mathbf{Z}_3^{k+1} = \mathrm{prox}_{\xi_3/\rho}\Big[\mathbf{A}^{\mathrm{H}}\Big\{\big(\mathbf{E}^{k+1}\big)^{\mathrm{H}}\mathbf{A}\mathbf{X}^k\Big\} - \mathbf{D}_3^k\Big] \tag{35}$$

where $\mathbf{D}_3$ is the dual variable denoting the residual for the variable splitting. Considering the complex-valued data, the sparse feature solution can be analytically derived as [29]

$$\begin{aligned} \mathbf{Z}_3^{k+1} &= \left[\left(\mathbf{W}_2{}^{k+1} - \frac{\xi_3/\rho\big(\mathbf{W}_2{}^{k+1}\big)}{|\mathbf{W}_2{}^{k+1}|}\right)\right]_+ \\ &= \frac{\max\big[\big(|\mathbf{W}_2{}^{k+1}| - \xi_3/\rho\big),0\big]}{\max\big[\big(|\mathbf{W}_2{}^{k+1}| - \xi_3/\rho\big),0\big] + \xi_3/\rho}\big(\mathbf{W}_2{}^{k+1}\big) \end{aligned} \tag{36}$$

where $\mathbf{W}_2{}^{k+1} = \mathbf{A}^{\mathrm{H}}\{(\mathbf{E}^{k+1})^{\mathrm{H}}\mathbf{A}\mathbf{X}^k\} - \mathbf{D}_3^k$. From (36), the complex soft-thresholding operator is a linear operator, which guarantees that the phase history of the data can be retentive. It ensures a fully embedded feature enhancement for the distributed local optimizations. The update function for the residual $\mathbf{D}_3$ can be given as

$$\mathbf{D}_3^{k+1} = -\mathbf{W}_2{}^{k+1} + \mathbf{Z}_3^{k+1} \tag{37}$$

which obviously denotes the difference between the primal variable $\mathbf{X}$ and the splitted variable $\mathbf{Z}_3$.

D. Global Optimization: Cooperative Learning

After the above three local optimizations finished, three feature variables $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ are splitted and updated in (25), (32), and (35), respectively. The structure feature, focusing feature, and sparse feature are accordingly solved. Besides, the dual variables $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$ measuring the residual errors for the variable splitting are developed in (26), (33), and (37), respectively. Then, the combined feature variable $\mathbf{Z}$ can be given as $\mathbf{Z}^{k+1} = [\mathbf{Z}_1^{k+1},\mathbf{Z}_2^{k+1},\mathbf{Z}_3^{k+1}]^{\mathrm{T}}$ and the combined dual variable $\mathbf{D}^{k+1} = [\mathbf{D}_1^{k+1},\mathbf{D}_2^{k+1},\mathbf{D}_3^{k+1}]^{\mathrm{T}}$. In such cases, the primal variable $\mathbf{X}$ can be updated in a global manner as

$$\begin{aligned} \mathbf{X}^{k+1} = \arg\min_{\mathbf{X}}\ &\frac{1}{2}\big\|\mathbf{Y} - \mathbf{E}\mathbf{A}\mathbf{X}^k\big\|_{\mathrm{F}}^2 \\ &+ \frac{\rho}{2}\big\|\mathbf{K}\mathbf{X}^k - \mathbf{P}\mathbf{Z}^{k+1} - \mathbf{D}^{k+1}\big\|_{\mathrm{F}}^2 \end{aligned} \tag{38}$$

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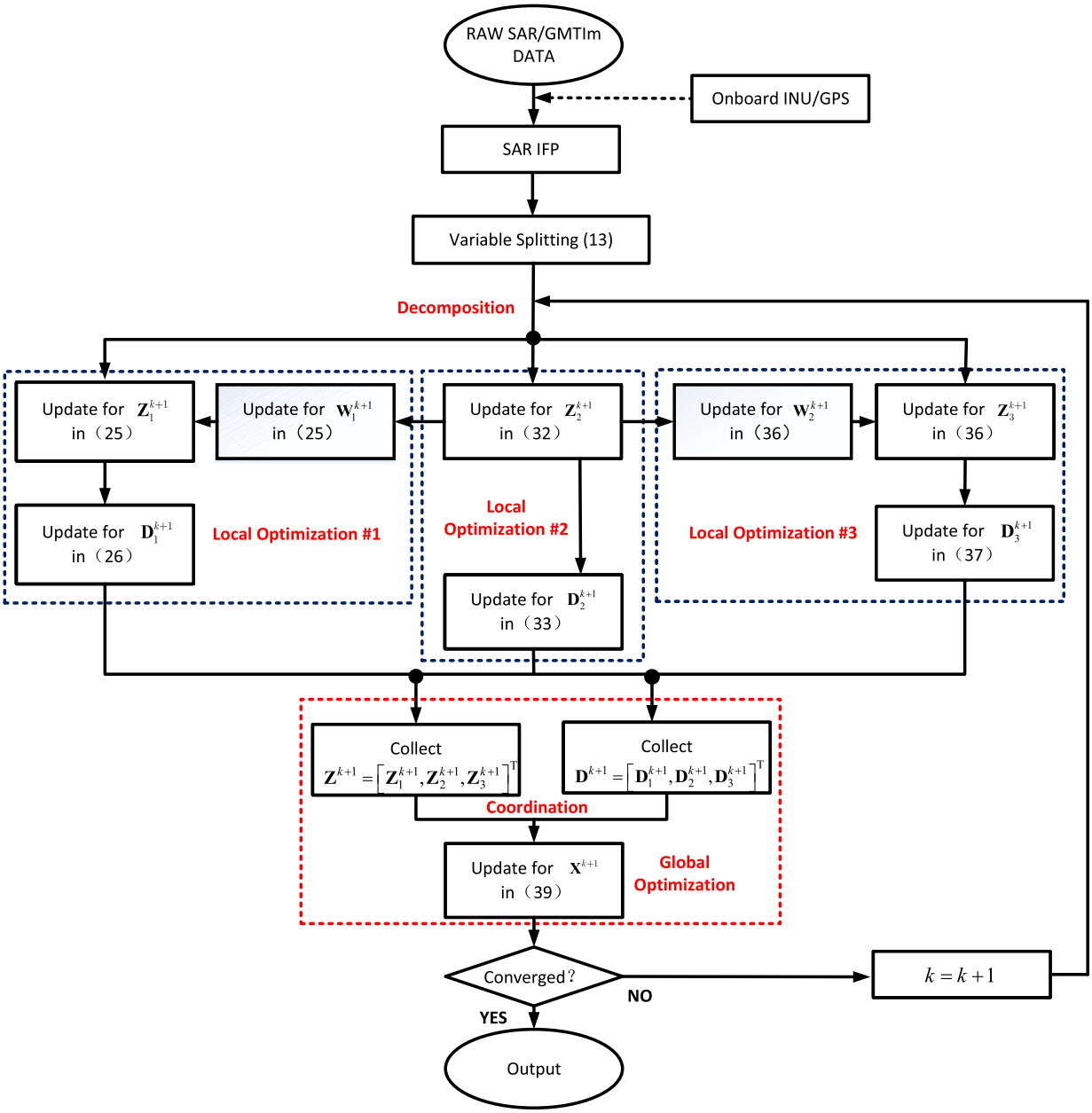


Fig. 3. Processing flowchart of the proposed SA-SAR algorithm.

where $\mathbf{K}=[\mathbf{J}_G,\mathbf{I},\mathbf{I}]^{\mathrm{T}}$ and $\mathbf{P}=\mathbf{I}$. Focusing on the objective function of (38), it is inherently a convex function, and a closed-form solution for $\mathbf{X}$ can be guaranteed as

$$\begin{aligned}\mathbf{X}^{k+1}=&\left(\mathbf{A}^{\mathrm{H}}\mathbf{A}+\rho\mathbf{J}_G^{\mathrm{H}}\mathbf{J}_G+2\rho\mathbf{I}\right)^{-1}\\&\cdot\Big[\mathbf{A}^{\mathrm{H}}\left(\mathbf{E}^{\mathrm{H}}\right)^{k+1}\mathbf{Y}+\rho_1\mathbf{J}_G^{\mathrm{H}}\left(\mathbf{Z}_1^{k+1}+\mathbf{D}_1^{k+1}\right)\\&+\rho_2\left(\mathbf{Z}_2^{k+1}+\mathbf{D}_2^{k+1}\right)\\&+\rho_3\left(\mathbf{Z}_3^{k+1}+\mathbf{D}_3^{k+1}\right)\Big]\end{aligned}\tag{39}$$

where ρ, ρ_1, ρ_2, and ρ_3 are for augmented Lagrangian parameters. From (39), the final feature enhanced SAR image $\mathbf{X}$ coordinates three feature variables $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ and three residual errors $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$. Since all the proximal operators are complex data compatible and they are linear operators, the coordination can be carried out in a cooperative manner between the three local optimizations and the global optimization.

V. Processing Procedure

Fig. 3 shows the processing flowchart of the proposed SA-SAR algorithm. Generally, it consists of four main steps of three local optimizations and one global optimization. The proposed algorithm is designed for raw SAR/GMTIm data processing. Prior to the first step, SAR IFP proceeds to correct the systematic range migration of stationary scene and moving targets [40]. Common algorithms, such as PFA, range-Doppler algorithm (RDA), and back-projection algorithm (BPA), are the candidates for the SAR IFP. If possible, the onboard

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INU/GPS can be utilized to compensate for the nonsystematic error to some extent [41]. Regarding GMTIm, proper preprocessings, including clutter suppression and parameter estimation, should be employed.

Next, under the ADMM mechanism, ℓ_1 norm, entropy norm, and STV norm are used to represent multiple features to be enhanced, and splitted variables are introduced according to the equivalence in (13). Setting $\mathbf{K}_1 = \mathbf{J}_G$, $\mathbf{K}_2 = \mathbf{I}$, and $\mathbf{K}_3 = \mathbf{I}$, the splitted variables corresponding to the structural feature, focusing feature, and sparse feature can be formulated, which are expressed as $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$, respectively. First, the structure feature enhancement is formulated in (22), where the STV regularizer is introduced for the prior of the clustering feature. Innovatively, the STV norm is derived in an analytical form of multiplication of Gaussian weighted Jacobain matrix and the intended image matrix as shown in (21). It paves a feasible way to perform the variable splitting for the structural variable, and the solution for the feature variable and the dual variable is developed in (25) and (26), respectively. Besides, to enhance the computational efficiency and convenience, the matrix trace and determinant are employed to replace the tedious eigendecomposition and/or singular decomposition.

Second, since the received SAR echo data contain inevitable phase errors, which will block the enhancement of sparse features and structural features, $\mathbf{Z}_2$ needs to be updated first. The focusing feature enhancement is modeled in (27) under the ADMM formulation, where the entropy norm is introduced for the prior. By employing a surrogate function, the focusing feature and the corresponding dual variable can easily be derived as shown in (32) and (33), respectively. To ensure fully cooperative learning among multiple local optimizations, the error calibrated data $\mathbf{W}_1$ and $\mathbf{W}_2$ in (25) and (36) are applied to update the structural feature and sparse feature, respectively. Third, ℓ_1 norm is introduced for the prior of sparse feature, and the solution for the feature variable and the dual variable is derived in (35) and (37), respectively.

Finally, the global optimization is performed in a cooperative manner, where the feature variables in (25), (32), and (35) and the dual variables (26), (33), and (37), are collected and coordinated in the combined feature variable $\mathbf{Z}$ and dual variable $\mathbf{D}$, respectively. Accordingly, the final SAR image $\mathbf{X}$ can be coordinately updated as shown in (39). Then, if a practical convergence condition is satisfied, the output of the proposed algorithm is an SAR image of scenes or targets of interest with enhanced structure, focusing, and sparse features simultaneously. The practical convergence condition can be the root-mean-square error between the current and the previous iterations. It can be noted from Fig. 3 that the procedure works in a "Decomposition-Coordination" mechanism, where the decomposition splits the joint optimization into three local optimizations, and the coordination merges the distributed features into a global variable for the intended SAR image.

VI. Experiments

In this section, experiments are carried out to verify the effectiveness of the proposed SA-SAR algorithm. Both simulated and raw SAR/GMTIm data are applied for verification.

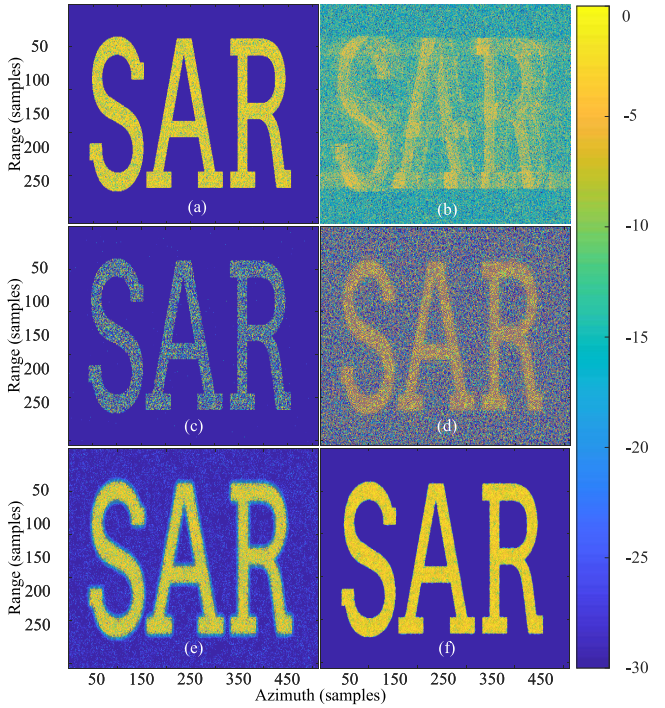


Fig. 4. Simulations. (a) Reference image. (b) Simulated image. (c) Lasso-ADMM. (d) TV-ADMM. (e) STV-ADMM. (f) Proposed SA-SAR.

Comparisons with conventional algorithms are performed, and the phase transition analysis [42] is utilized to show the superiority of the proposed algorithm in terms of both quantitative and qualitative aspects.

A. Simulated Data Experiments

First, a synthesized simulation is carried out, and the results are shown in Fig. 4. In order to reflect the structural characteristics, defining a fixed area with the shape in the characters "S," "A," and "R," totally, 803 simulated scatterers are randomly distributed to form the "SAR" shape, and each scatterer is assumed isotropic. The target scenario has a dimension of 256 × 512 (range × azimuth). The radar is simulated in X-band with centroid frequency of 9.75 GHz, bandwidth of 1.2 GHz, airborne platform height of 6000 m, and pulse repetition frequency (PRF) of 1200 Hz. The reference image is shown in Fig. 4(a). Transforming the simulated data to the range compression domain, adding the Gaussian white noise (SNR = 0 dB), and multiplying a phase error function with quadratic distribution in 100-rad dynamic range to each range cell of the data, the Fourier transform is applied to the echo data to obtain the RD image, as shown in Fig. 4(b). Fig. 4(c)–(f) shows the results by Lasso-ADMM [43], TV-ADMM [39], STV-ADMM [27], and the proposed SA-SAR, respectively, where Lasso-ADMM, TV-ADMM, and STV-ADMM are the processing algorithms with ℓ_1, TV, and STV regularizations, respectively. Conventional minimum entropy autofocusing is applied for the phase error autofocusing [44] in Lasso-ADMM, TV-ADMM, and STV-ADMM in a tandem manner. Visually, the imaging result by the proposed SA-SAR algorithm performs the best, where it almost recovers the same ground truth as in Fig. 4(a). The correlation (COR)

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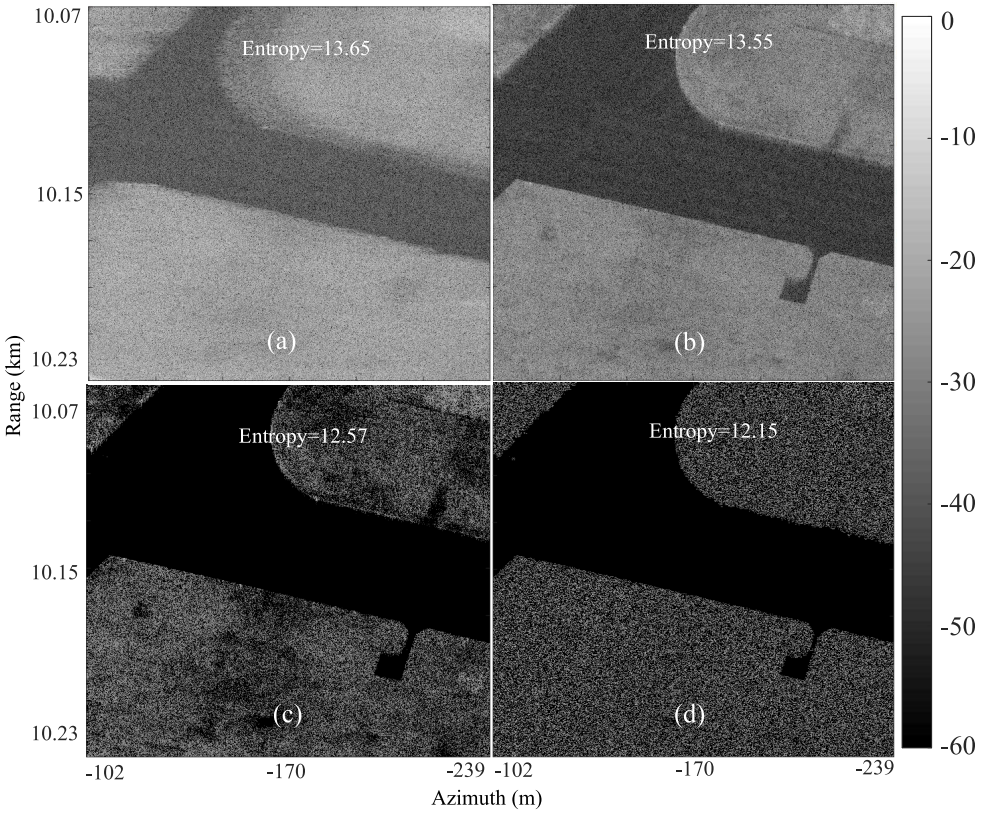


Fig. 5. Comparison of multifeature enhancement (a) without feature enhancement, (b) with focusing feature enhancement, (c) with focusing and sparse feature enhancement, and (d) with multifeature enhancement by the proposed SA-SAR.

between the experimental result and the reference image and the related root mean square error (RMSE) can be used for a quantitative evaluation. The RMSE by SA-SAR in Fig. 4(f) is 2.6149 and COR is 0.9315. The RMSE by Lasso-ADMM in Fig. 4(c) is 3.8136 and COR is 0.8868. Besides, the RMSEs by TV-ADMM and STV-SAR in Fig. 4(d) and (e) are 3.6954 and 2.9650, and the CORs are 0.8879 and 0.9245, respectively. From the comparisons, the proposed SA-SAR achieves the smallest RMSE and the highest COR. Since the reference image exhibits an obvious structural ground truth, the proposed SA-SAR is capable of reaching the highest accuracy in recovering the structure feature. Also, it calibrates the phase error in a parallel and cooperative manner. Regarding computational complexity, the proposed algorithm developed within the ADMM framework is superior compared with the conventional convex optimization and Bayesian learning framework. For the simulation data in the size of N and M in number of pulses and range cells, the complexity of the proposed SA-SAR algorithm is in the order of $O(2M \log N)$ [28], the conventional convex optimization (CVX) is in the order of $O(MN^2)$ [15], and the conventional SBL is in the order of $O(MN \log N)$ [15]. Obviously, the proposed algorithm is the best in terms of computational complexity.

B. Raw SAR/GMTIm Data Experiments

Next, the experiments on raw SAR data are carried out, and the results are shown in Fig. 5. The raw SAR data were collected by an airborne X-band radar in the spotlight mode. The operational range to the observing scene center is about 10 km, the flying speed of the airborne platform is ∼150 m/s, the PRF is 2100 Hz, and the transmitting bandwidth is ∼1.2 GHz. Because the navigational data are unavailable, the SAR image by the common IFP is defocused, as shown in Fig. 5(a). To this end, the data-driven error autocalibration is necessary. By employing the conventional error compensation [44], the refocused SAR image is given in

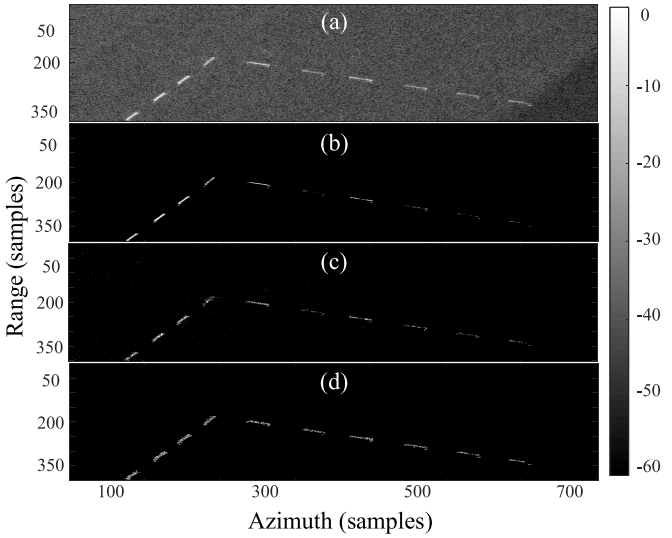


Fig. 6. Results of Sandia data set 1. (a) RD. (b) Lasso-ADMM. (c) TV-ADMM. (d) SA-SAR.

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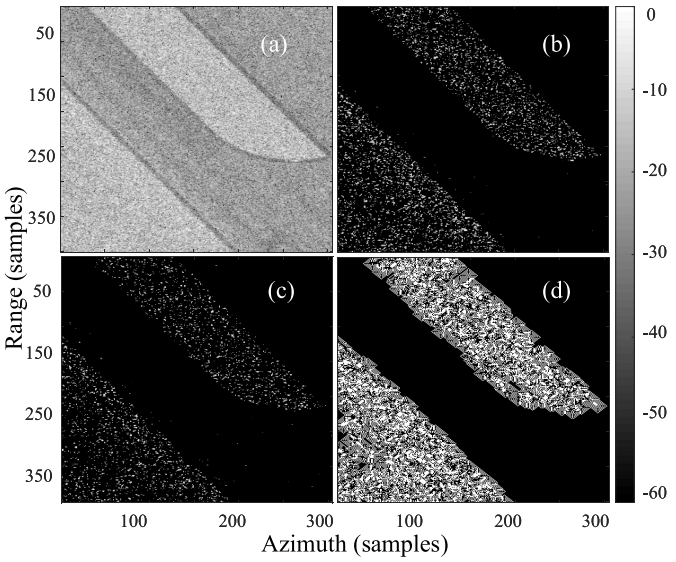


Fig. 7. Results of Sandia data set 2. (a) RD. (b) Lasso-ADMM. (c) TV-ADMM. (d) SA-SAR.

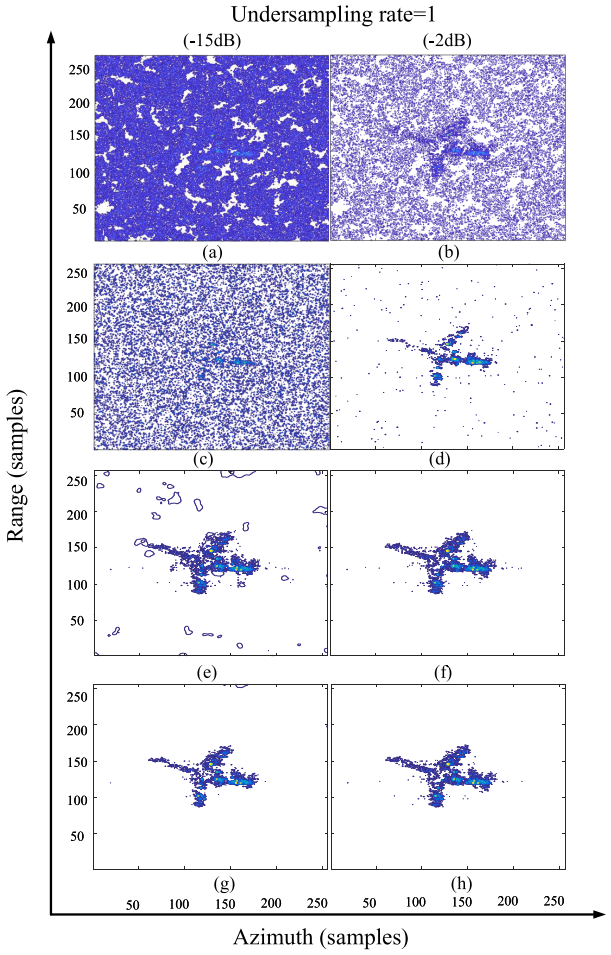


Fig. 8. Comparisons under different SNRs without undersampling. (a) and (b) TV-ADMM. (c) and (d) Lasso-ADMM. (e) and (f) STV-ADMM. (g) and (h) SA-SAR.

Fig. 5(b), where the focusing quality improves to some extent. In Fig. 5(c), the imaging result by the error autocalibrated sparse feature enhancement [29] is demonstrated. Although both the features are highlighted and the image contrast

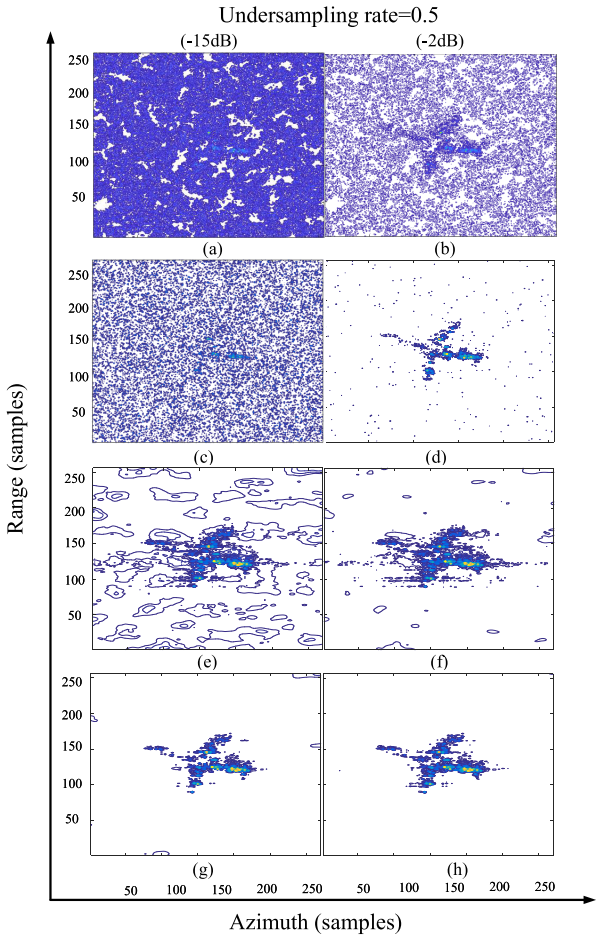


Fig. 9. Comparison under different SNRs with the USR 0.5. (a) and (b) TV-ADMM. (c) and (d) Lasso-ADMM. (e) and (f) STV-ADMM. (g) and (h) SA-SAR.

improves greatly, the structural features are not apparent, such as the grass land, the borderline, and the spacing area in the grass land. In Fig. 5(d), the proposed SA-SAR algorithm proceeds, and the resultant image achieves the best performance in terms of focusing quality, image contrast, and grass-land shape. From the comparison between Fig. 5(c) and (d), both the algorithms achieve satisfied focusing quality, but the image content in Fig. 5(c) is nonhomogeneous, e.g., the grass land in Fig. 5(c), which may be the consequence of the sparsity enhancement. The proposed SA-SAR algorithm coordinates the sparse, focusing, and structural features in a cooperative manner, and it overcomes the shortcoming of the conventional algorithms.

To further verify the applicability of the proposed SA-SAR algorithm for various structures, two additional data sets are employed. The data sets are Ka-band and with the resolution up to 0.1 m. The experimental results for the first data set are shown in Fig. 6, where the RD, Lasso-ADMM, TV-ADMM, and the proposed SA-SAR are applied. From the comparison, the ground truth is with multiple steel tubes for isolation, and the RD image in Fig. 6(a) distinguishes the tubes, while the background clutter is high. The Lasso-ADMM achieves the sparse enhancement, and the target-to-clutter ratio improves greatly, as shown in Fig. 6(b). However, some linear features are lost due to the sparse prior incorporated. The conventional

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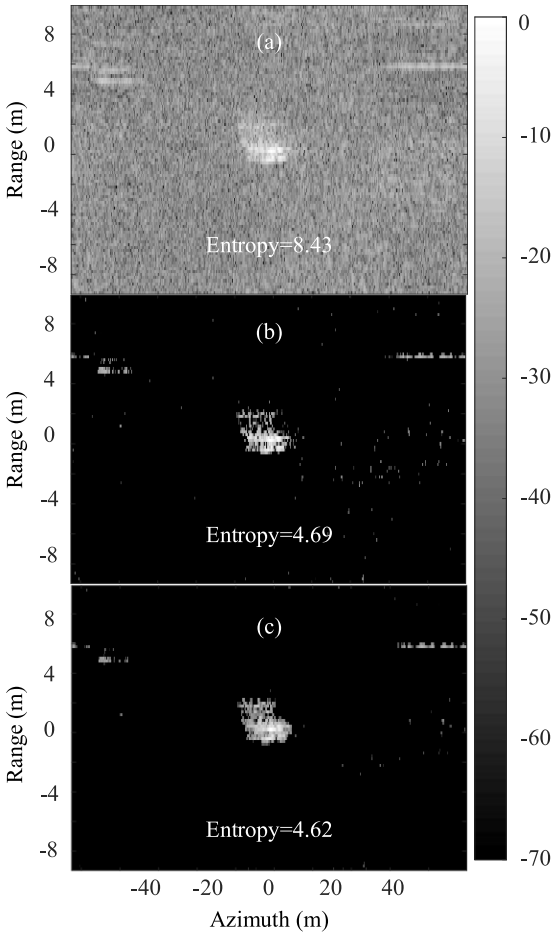


Fig. 10. Resultant of Durango images. (a) RD. (b) Lasso-ADMM. (c) SA-SAR.

TV-ADMM guarantees the linear structure features to some extent in Fig. 6(c). However, the result by the proposed SA-SAR achieves the best performance visually. A high contrast image can be obtained, and all the metal barriers in linear structure can be obviously distinguished.

The second data set is with rich features of planar surface structure. The experimental results are shown in Fig. 7, where the RD, Lasso-ADMM, TV-ADMM, and the proposed SA-SAR are applied. A similar conclusion can be observed that the proposed algorithm achieves the best performance in terms of enhancing the structural features.

Furthermore, the Yak-42 ISAR data are used for the verification of the proposed algorithm. Various SNR and undersampling rate (USR) are simulated, and Lasso-ADMM, TV-ADMM, STV-ADMM, and SA-SAR algorithms are employed for the comparison. The experimental results are in Figs. 8 and 9. In Fig. 8, no undersampling is applied, and both SNR of -15 and -2 dB are simulated. From the comparison, the proposed SA-SAR algorithm is capable of recovering the most complete structure feature of an airplane, and the background noise is suppressed at the lowest level. However, when only STV is incorporated within the ADMM, the resultant image is with a relatively high noise level. In Fig. 8, the echoed data are undersampled in 0.5. Due to the undersampled data, the imaging results are degraded, but

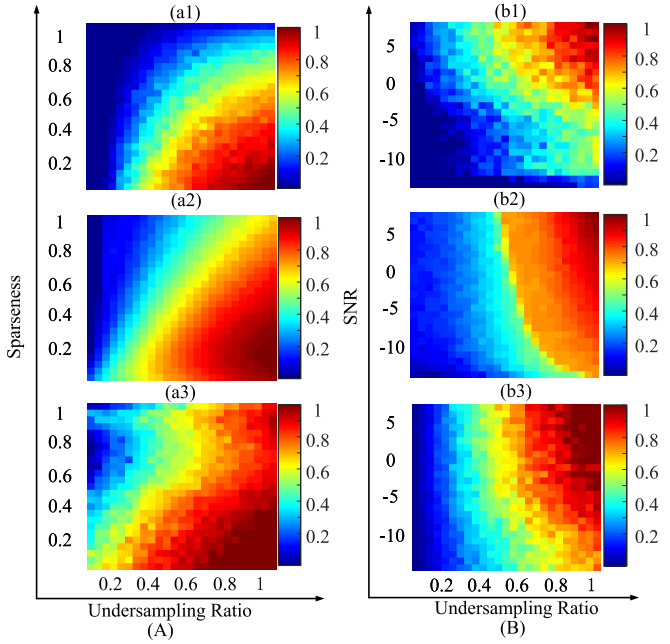


Fig. 11. PTD. (a1) and (b1) TV-ADMM. (a2) and (b2) Lasso-ADMM. (a3) and (b3) SA-SAR.

the proposed algorithm is still capable of highlighting the structure feature with the best accuracy.

Finally, the raw GMTIm data set named by Gotcha is applied to examine the performance of the proposed algorithm for GMTIm. From the above analysis, it is obviously that the TV norm is not applicable to recover the plane-surface structure. Therefore, comparisons among the conventional RD, Lasso-ADMM, and the proposed SA-SAR algorithm are carried out in this experiment. The Gotcha data set is published by the US Air Force Research Laboratory (AFRL) in 2010, which was collected by an airborne SAR in X-band, PRF in $\sim$2100 Hz, operational range in $\sim$10 km, bandwidth in 640 MHz. A cooperative moving target named Durango is within the radar beamwidth, and it is a cooperative target moving with a recorded velocity. Due to the target velocity, the moving target response will be displaced and defocused in the SAR image. The RD algorithm and Lasso-ADMM can hardly meet the refocusing performance, and the autofocusing algorithm should first be used to refocus the imaging results. Fig. 10(a) and (b) shows the experimental results by RD and Lasso-ADMM algorithms, respectively. The Durango image by the RD algorithm encounters severe clutter from the background. While the Lasso-ADMM achieves sparse feature enhancement, accordingly, the clutter is suppressed to a great extent. However, the enhancement of sparse feature easily loss structural features of the target of interest. Fig. 10(c) shows the experimental result by the proposed SA-SAR algorithm, which has the cooperative enhancement ability of structural, sparse, and focusing features. It obtains the best performance in terms of high SNR and structure recovery.

C. Performance Evaluations

To examine the performance of the proposed algorithm from the quantitative aspect, the phase transition analysis [42] is

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performed. The phase transition analysis is to examine the performance of the recovering algorithm in terms of various conditions. In Fig. 11, different conditions in terms of sparseness, undersampling ratio, and SNR are simulated, and the left column results are the phase transition diagrams (PTDs) in terms of sparseness and undersampling ratio by TV-ADMM [39], Lasso-ADMM [43], and SA-SAR, respectively. The right column is that in terms of SNR and undersampling, respectively. For each specific condition, 500 independent and identical Monte Carlo experiments are carried out, and a mean value of the correlation metric is calculated. The larger the correlation, the higher accuracy of the structure feature enhancement. From the comparison, the area with large correlation by the proposed SA-SAR is the biggest, which means that the proposed algorithm is with the best performance in terms of robustness in the structure feature recovery.

VII. Conclusion

In this article, a novel SA-SAR algorithm, which combines STV, entropy, and ℓ_1 norms within a cooperative ADMM framework, is proposed to achieve multiple-feature enhancement and overcome the loss of structural feature by conventional sparsity-driven algorithms. The proposed algorithm works in a "Decomposition-Coordination" procedure, where the variable splitting scheme is utilized for the decomposition of joint feature enhancement. The structure feature, focusing feature, and sparse feature are solved in local minimizations, and they are finally coordinated in the global optimization. By encoding the STV, entropy, and ℓ_1 norm into the multitask learning framework, the ADMM method is capable of achieving a balanced optimization for a global optimum. To facilitate the incorporation of the STV norm, the Jacobian matrix is introduced, and the matrix trace and determinant are utilized to improve the computational efficiency of the structure feature enhancement. Besides, a surrogate function is designed to replace the original objective function for the entropy optimization, and a closed-form solution for the focusing feature can be guaranteed. Both simulated and raw SAR/GMTIm data are applied for the verification of the proposed SA-SAR algorithm, and the classic phase transition analysis is employed to verify the performance quantitatively.

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